



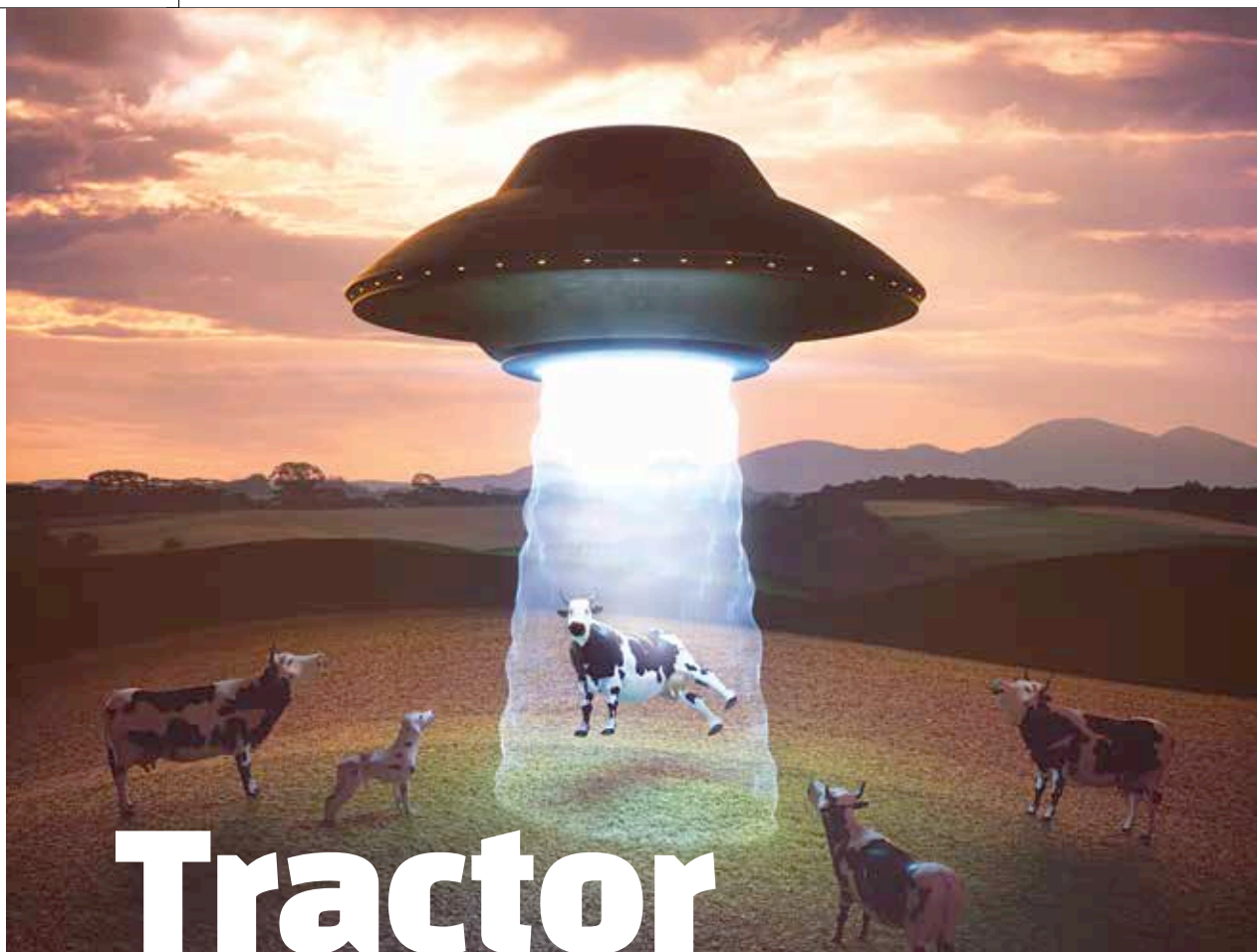
Exercise for Materials

Mechanical training makes hydrogels strong and fatigue resistant.
<https://digit.in/may19-75>



Snake robot slithers better

New version of robotic snake inspired by kirigami is better than its predecessor.
<https://digit.in/may19-76>



Tractor Beams

Rays or waves that grab and move objects has been a science fiction staple, which has recently been defictionalised.

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Tractor beams have been a staple of science fiction, and we have seen them in both *Star Trek* and

Star Wars. There are many approaches to realise the technology in real life, but so far the successful demonstrations have been only on micrometer

scale particles, and not for abducting farm animals as we've shown above.

OPTICAL BEAMS

An optical vortex made up of counterpropagating beams of light could trap objects in the dark center. The trapped object can be moved along the center by alternatively strengthening or weakening one of the beams of light, heating up the air molecules around the material. Researchers at the Australian National University have demonstrated such an approach to manipulate particles of gold coated hollow glass. After the particles are trapped in the dark centers, energy from the laser heats up hotspots on their surface. The air molecules strik-

ing these hotspots shoot away from the surface, propelling the particles in the desired direction. The generally toroid shape of the laser beam is changed from say a star shaped polarization to a ring shaped one, to manipulate the exact positions of the hotspots, allowing the researchers to move the particles towards or away along the axis of propagation.

Another approach is to use an optical solenoid beam, the intensity of which peaks in spirals along the axis of propagation. In 2018, researchers from Harvard demonstrated such an approach, using what was essentially an optical version of the Archimedes Screw, used to draw water. Depending on the direction of rotation of the heli-